

SYED DANISH ALI

[LinkedIn](#) | [GitHub](#) | [Website](#) | [Email](#)

PROFILE

- Early-career researcher with experience in glacier & climate dynamics, empirical-statistical downscaling & machine learning approaches.
- Interested in abrupt Earth-system changes, paleoclimate reconstruction, proxy-based analysis, & climate modelling.

EDUCATION

Master of Science in Marine Environment & Resources

Sep '23 - Sep '25

Erasmus Mundus Joint Master's Degree (MER2030)

Europe, EU

- **Université de Bordeaux** (France), **Universidad del País Vasco** (Spain), **Université de Liège** (Belgium)
- GPA: 8.31 / 10 – *High honors*
- Nominated finalist – Prof. M.B. Collins Award for Best Master's Thesis (2025)
- Erasmus+ Scholarship & Grants

Bachelor of Science (Honors) in Geology

Jul '20 - Jul '23

University of Delhi

New Delhi, India

- GPA: 8.122 / 10 GPA – *First Division*
- Minor in Mathematics & Physics
- L. Austin Week's Undergraduate Grant (by American Association of Petroleum Geologists- U.S.A.)

RESEARCH EXPERIENCE

Visiting Researcher

Feb '25 – Sep '25

University of Glasgow: School of Geographical and Earth Sciences ([Climate Dynamics Lab](#))

Scotland, U.K.

- **Master Thesis** “Detection and explanation of shifts in the climatic controls on Norwegian glaciers” – supervised by Dr. rer. nat. habil. Sebastian G. Mutz.
- **Web-Links:** [Thesis Manuscript](#) | [Presentation Recording](#)
- **Key Competencies:**
 - Designing and executing the research plan in collaboration with the supervisor.
 - Working with the ‘pyESD’ package in Python to adapt and extend the package for glacier applications.
- Drafting a peer-review-style manuscript of the thesis findings for submission to a Q1 journal.

Research Intern (PlasteMER project)

Jul '24 - Aug '24

Universidad del Pais Vasco (UPV/EHU)

Plentzia, Spain

- Focused on the assessment of microplastic (MP) abundance in the Urdaibai Estuary, Basque Country.
- **Key Competencies:**
 - Assisting a PhD student involved in the project.
 - Operating a laminar-flow cabinet in a clean laboratory to minimize contamination.
 - Performing the chemical digestion of polychaete samples to quantify MP abundance.
 - Preparing laboratory reagents (e.g. acetic acid, ethanol, filtration membranes).

Research Intern (ECOTRANSEAS project)

May '24 - Jul '24

Universidad del Pais Vasco (UPV/EHU)

Plentzia, Spain

- A Professional Placement in a European project focused on the ecotoxicological analysis of ballast water from 3 ports in Spain (Algeciras, Bilbao, & Valencia).
- **Key Competencies:**
 - Categorizing and organizing water samples from the ports.
 - Measuring and analyzing the chemical properties of the water samples.
 - Conducting toxicity tests on sea-urchin embryos (SET, ICES, 2012) & polychaetes (coelomocytes viability).

OTHER EXPERIENCE

Climate Data Analyst

BETWA (Bettering Environmental Technology and Water Advisory)

Feb '26 – Mar '26

New Delhi, India

Project: *Spring-shed Management & Water Security in Yakten, Sikkim, Himalayas.*

- Analysing large climate datasets to extract climatological baselines, long-term trends, and extreme variability using threshold-based and distributional diagnostics.
- Integrating spatial and seasonal climate variability analyses to quantify recharge-relevant hydro-climatic dynamics and support policy intervention.

Student Help Team: Library Manager (Part-Time)

Universidad del Pais Vasco (UPV/EHU)

Apr '24 - Jun '24

Plentzia, Spain

- Management of the research archive at the Plentzia Marine Station.
- Responsibilities included cataloging & organizing the research paper archive, maintaining and preserving records, facilitating access, ensuring compliance with guidelines, and supporting research activities.

PROJECTS

Ecological Modeling – NPZDO₂ Biogeochemical Model for eutrophication management, nutrient speciation, & sediment interactions. Data Analysis & Visualization in R (used FME for parameter sensitivity).

2D Diffusion Model in Geophysics: Simulation of pollution patch using Finite Difference Method (FDM). Programming with Julia to test diffusion models using sparse matrices and iterative solvers for efficiency.

Sertraline Toxicity in Copepods (*Acartia tonsa*): Designed an acute toxicity bioassay–Sertraline (SRT) on marine copepods over 24h and 48h exposure. Conducted ANOVA, Probit analysis, and time-dependent mortality tests using R and IBM SPSS.

Estuarine Hydro-Sedimentary Model- Gironde Estuary, France: Processed ADCP velocity profiles, CTD data, and LANDSAT-8 imagery to map suspended particulate matter and turbidity. Used bottom-tracking ADCP methods to quantify sediment transport and turbulence during tidal reversals in the Gironde Estuary.

Ecological Quality Assessment of Intertidal Communities in Bilbao, Spain: Applied PRIMER and PERMANOVA to assess macrobenthic community structure and pollution gradients in intertidal zones of Bizkaia, Spain, and identified long-term ecological recovery trends through historical data comparison.

SKILLS

- **Technical:**
 - Programming: Python, R, Julia
 - Typesetting: LaTeX
 - GIS: QGIS & ArcGIS, SNAP (Sentinel Application Platform)
 - IBM SPSS Statistics
 - Microsoft Office Suite: Excel, Word, Access, etc.
 - Mathematica (Wolfram), Circuit designing (LT Spice)
 - Adobe: Photoshop, Illustrator, AI
- **Language:**
 - English: C2
 - French: A1
 - Hindi: C2
 - Urdu: C1
- **Other:**
 - International Driving License Permit

REFERENCES

1. **Name:** Dr. rer. nat. habil. Sebastian G. Mutz
Capacity: Master Thesis Supervisor
Institute: University of Glasgow, Scotland (UK)
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2. **Name:** Prof. Manuel Soto
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